

Water Automation Schematic Drawings

4 clean technical diagrams. Pure Python. No AI.

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Top-down house, every cable run color-coded

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COMPANION DOCS

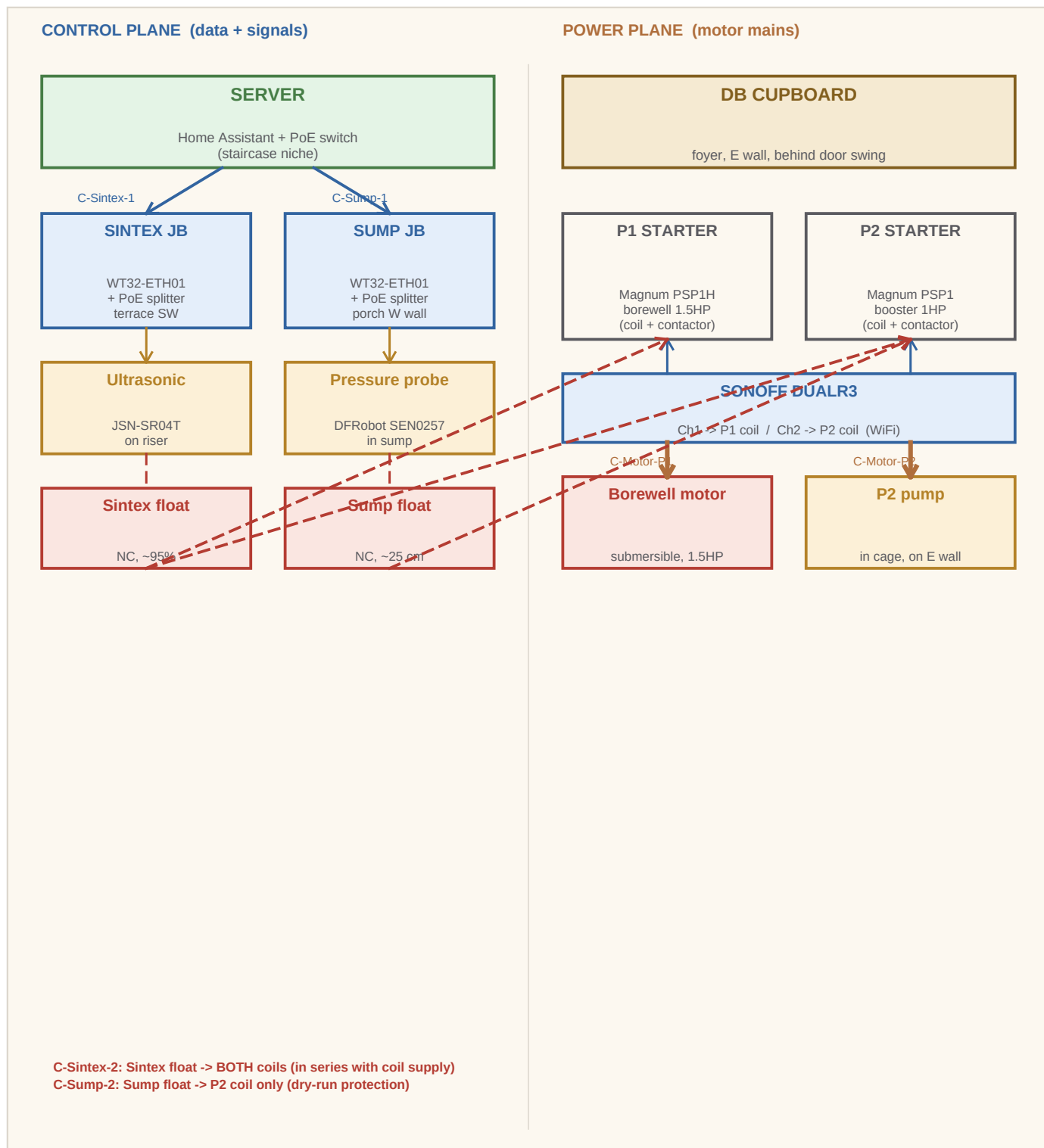
WATER_AUTOMATION.pdf - full long-form spec (11 pages)

WATER_AUTOMATION_VISUAL.pdf - AI-rendered scene visualisations

v1.0 - 2026-05-29 - Pure-Python schematic deck

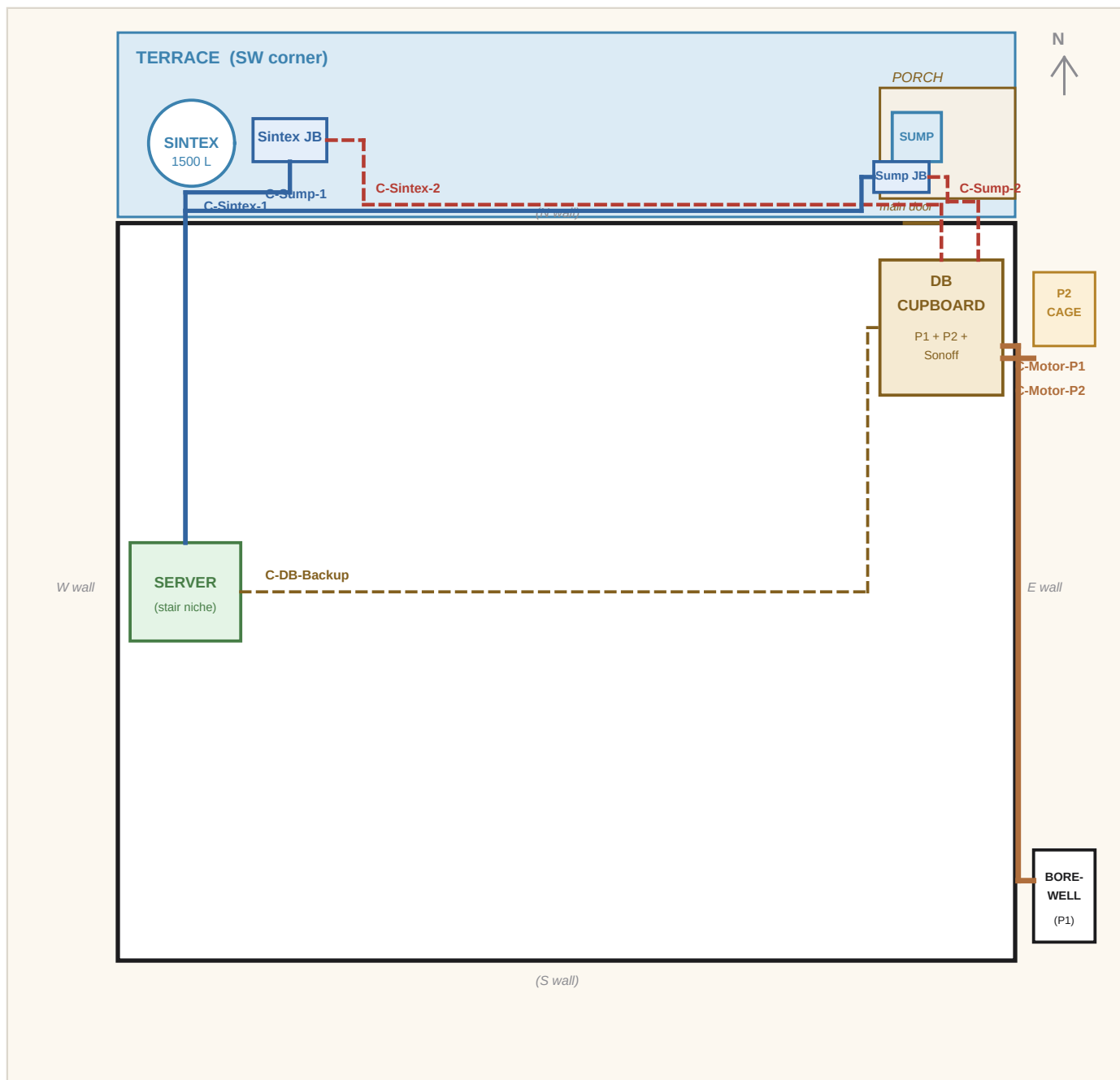
System Block Diagram

Who talks to whom



Floor plan + conduits

Top-down, all 7 runs colour-coded

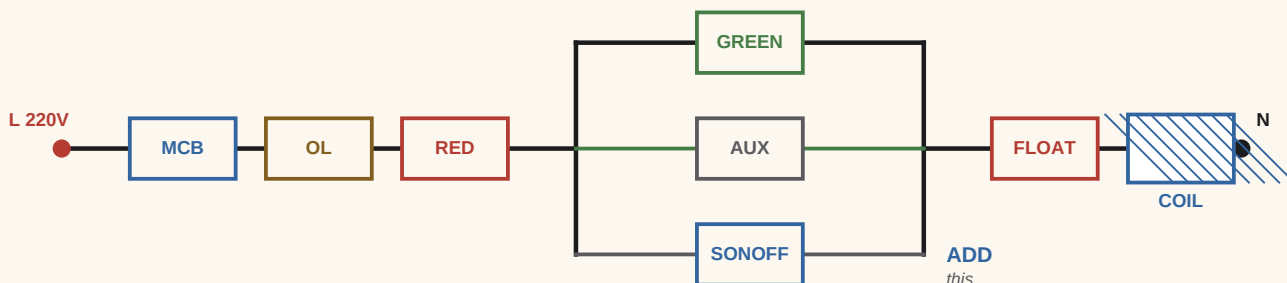


Starter wiring

How the Sonoff parallel-taps the Magnum Pradhaan

MAGNUM PRADHAAN CONTROL CIRCUIT + SONOFF PARALLEL TAP

Identical for both P1 (1.5HP) and P2 (1HP). Manual buttons remain functional.



LEGEND: **MCB** = 16A Type C **OL** = thermal overload **RED** = manual stop (NC) **FLOAT** = Sintex high (NC) **COIL** = contactor coil
GREEN = manual start (NO) **AUX** = latching (NO) **SONOFF** = server start/stop (NO)

Existing path: L -> MCB -> OL -> RED(NC) -> [GREEN latched by AUX] -> FLOAT(NC) -> COIL -> N
 Sonoff is wired in PARALLEL with the GREEN button. Manual + server both can start the motor.

WHAT THIS BUYS US

Server can start/stop the motor via Sonoff WiFi. Manual operation still works (green/red buttons unchanged). Thermal overload (OL) and Sintex high-float (FLOAT) remain in series with the coil - true hardware safety, independent of any server.

WIRING RULE

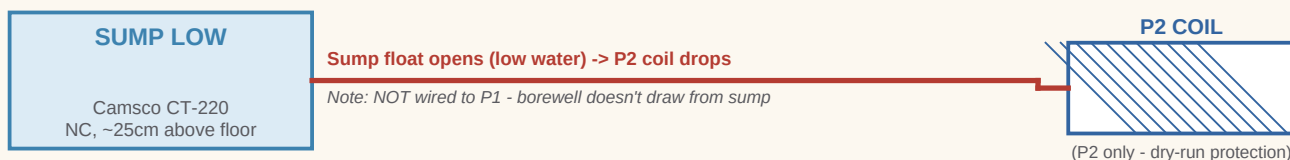
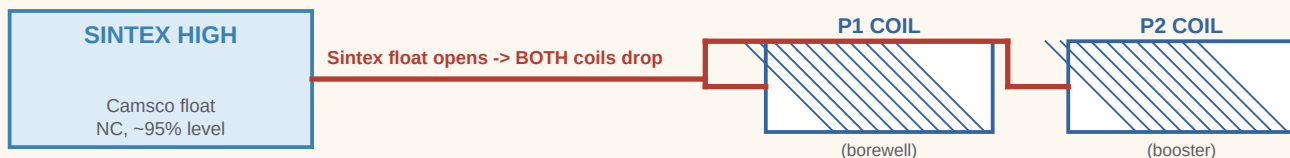
Sonoff's switched output goes in PARALLEL with the green button - NEVER in series, NEVER bypasses the float. Test manual operation BEFORE energising the Sonoff for the first time.

Float failsafe topology

Mechanical safety: how floats interrupt the contactor coils

FLOAT FAILSAFE WIRING - SERIES TOPOLOGY

Both floats are NC (normally closed). When water triggers the float, the contact *OPENS*, breaking the coil supply.



These wires are 220 V mains-class - 2-core 1.5 sqmm double-insulated, in separate 16 mm conduits (C-Sintex-2 and C-Sump-2). NEVER share conduit with Cat6.

WHY BOTH FLOATS ARE MECHANICAL, NOT SERVER-CONTROLLED

Mains-voltage NC contacts in series with the coil supply means the failsafe is physical and instant. No CPU, no firmware, no WiFi can fail it. If the entire smart system burns down, the floats still cut the motor when water reaches the trigger level.